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Stacking-Angle-Manipulated Emission (SAME) in Anthracene-Based Figure-Eight Metallo-Supramolecules

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ABSTRACT

Stacking angles played a decisive role in the coupling strength of the excited state, the overlap of electronic orbitals, and behavior of excitons, which further have ultimately affected the luminescent properties. However, developing effective strategies to precisely tailor molecular stacking angles of chromophores still remains challenges. In this work, we constructed a series of figure-eight supramolecules **S1–S3** through the coordination-driven self-assembly of anthracene-based 180° di-platinum(II) acceptor **L** and ditopic pyridyl ligands **L1–L3**, respectively. Variation in ligand length enabled regulation of intramolecular anthracene stacking angles in the assembled structures and photoluminescent properties. Photophysical studies revealed that larger stacking angles significantly enhance fluorescent intensities and photoluminescence quantum yields in both solution and solid states. Femtosecond transient absorption spectroscopy further demonstrated that the excited-state lifetimes of **S1–S3** were extended due to suppressed non-radiative decay pathways. Moreover, density functional theory calculations showed that the increasing stacking angles weakened intramolecular anthracene interactions, leading to enhanced radiative transition rates. This study elucidated the relationship of molecular packing and luminescent properties, which will pave the way for construction of materials with excellent luminescent performance.

1 | Introduction

Luminescent materials have attracted considerable attention due to their wide-ranging applications [1–7] in light-emitting diodes (LEDs) [8, 9], biological imaging [10–12], environmental sensing [13, 14], probes [15], semiconductors [16–18], and other fields [19–21]. Among the factors influencing luminescent performance, the packing modes and spatial arrangements of chromophores play a critical role in determining their photophysical properties [22–26]. Consequently, precise control over chromophore packing has become the focus and breakthrough points in establishing the relationship between structural design and functionality [27–32]. Therefore, developing effective strategies to modulate molecular

packing at the nanoscale is essential for tuning luminescent properties and deepening understanding of the structure–property relationship [33, 34].

Over the past decades, chemists have devoted significant efforts to tuning the molecular packing of chromophores and their photoluminescent properties [27, 33, 35–43]. Several strategies (Scheme 1a) have been developed to control packing modes of fluorophores, such as extension of π -conjugation [44], incorporation of heteroatom in conjugated backbone [45], cocrystal [46, 47], and introduction of steric motifs [48–50]. Among the factors affecting packing modes, stacking angles played a decisive role in the excited-state coupling strength, electronic orbitals overlap,

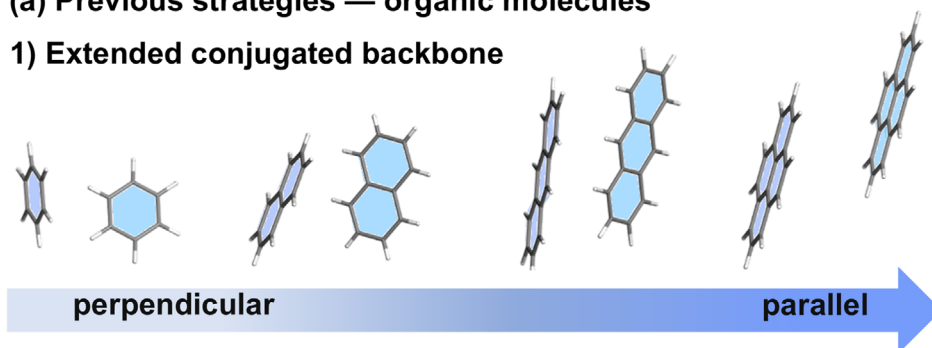
Junjuan Shi and Yunting Zeng contributed equally to this work.

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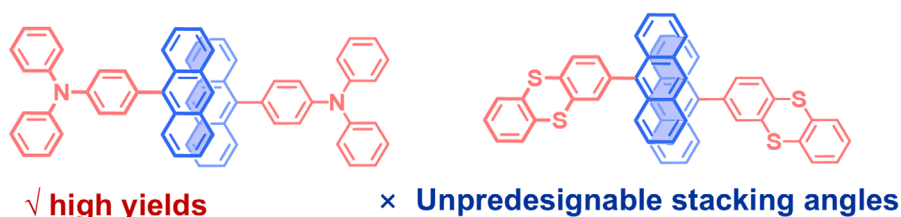
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(a) Previous strategies — organic molecules

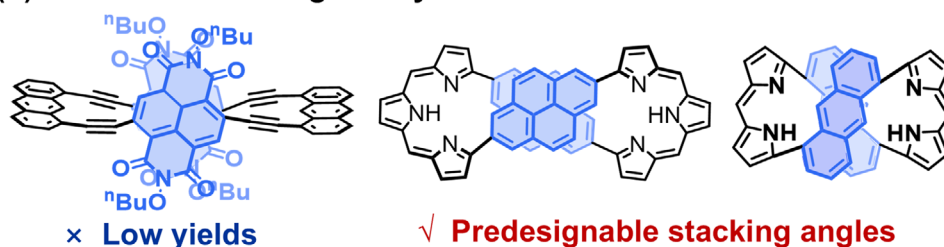
1) Extended conjugated backbone



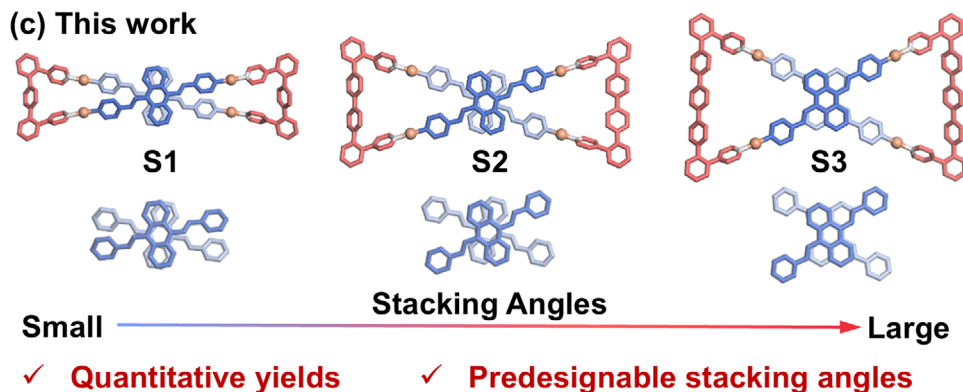
2) Modification with steric motifs



(b) Construction of organic cycles



(c) This work



SCHEME 1 | Previous construction strategies (a, b) and our strategy (c) to tailor the interaction of chromophores.

and exciton behavior, thereby further influencing on the luminescence [51, 52]. However, it remains difficult for these methods to effectively tune the stacking angles of chromophores. To achieve predictable and programmable control over chromophore interactions, chemists have endeavored in construction of twisted covalent cycles with rigorous molecular design (Scheme 1b) [53–58], yet which are low yield, time-consuming, and irreversible synthesis. Thus, developing a customizable and effective strategy to tailor stacking angles and establishing clear relationships between chromophores arrangements and luminescent modulation remains a formidable challenge.

Herein, exploiting the reversible nature of metal–ligand interactions [59–68], we designed and constructed a series

of figure-eight supramolecules **S1–S3** by coordinating an anthracene-based 180° di-platinum(II) acceptor **L** and ditopic pyridyl ligands (**L1–L3**) with different lengths in quantitative yields, respectively (Scheme 1c). Owing to the deliberate design of ligands with different lengths, the stacking angles between chromophores in these assemblies could be finely tuned in a predesignable manner. Photoluminescence spectra revealed that both photoluminescence quantum yields (PLQYs) and lifetimes are improved with the increasing stacking angles of **S1–S3**, which are higher than these of ligand **L**. And the ratio of non-radiative decay rate (k_{nr}) and radiative decay rate (k_r) of **S3** is 26.35, which is far lower than that of **S1**. These results indicated that larger stacking angles of chromophores are more effective in suppressing non-radiative decay pathways. Density

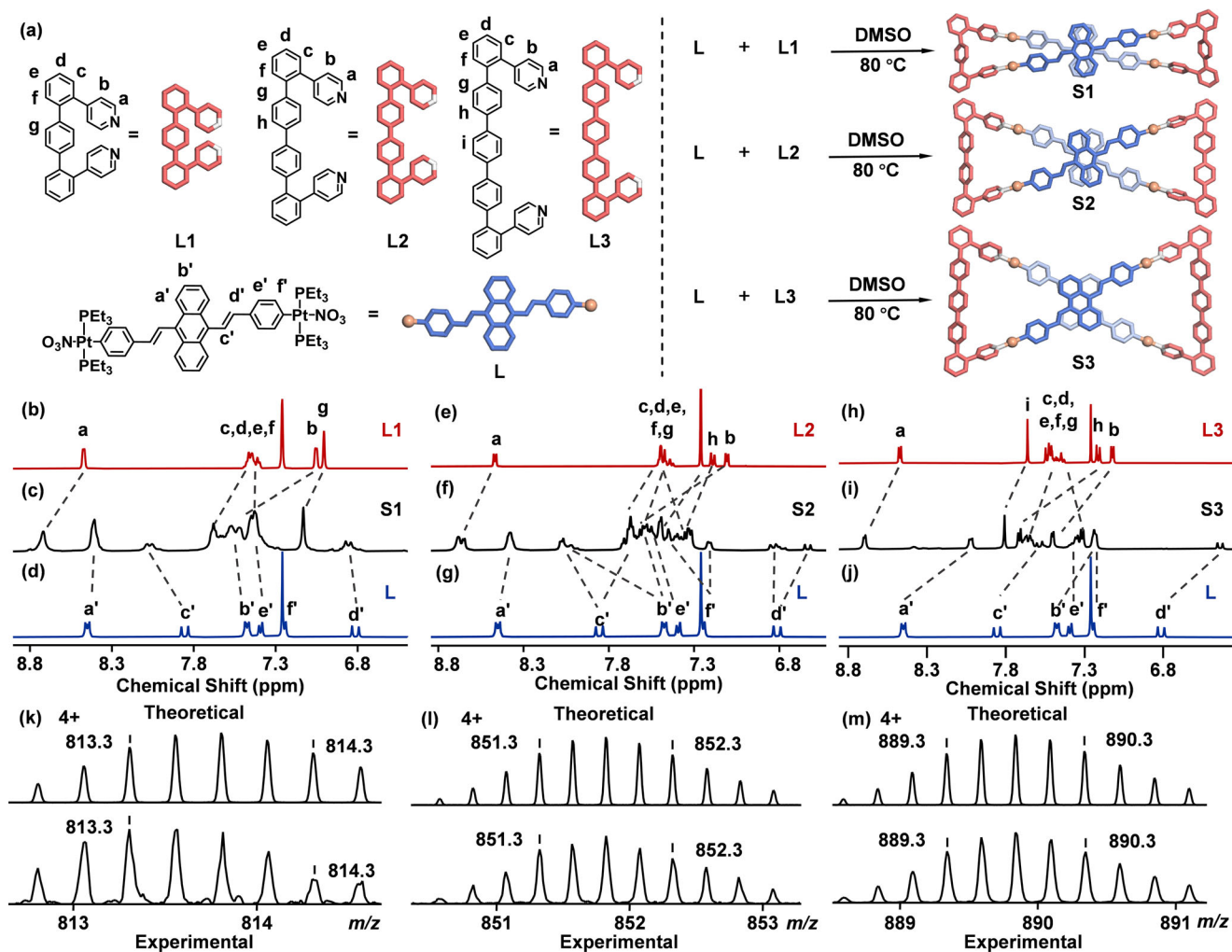


FIGURE 1 | Self-assembly of complexes **S1**–**S3** (a); ¹H NMR spectra (500 MHz, 300 K) of **L1** (b), **L2** (e), **L3** (h), and **L** (d, g, j) in CDCl₃ and **S1** (c), **S2** (f), and **S3** (i) in DMSO-*d*₆; isotope patterns of **S1** (k), **S2** (l), and **S3** (m).

functional theory (DFT) calculations showed that intramolecular interactions of anthracene motifs decreased from **S1** to **S3**, leading to enhance radiative transition rate, thereby increase the fluorescent intensity, and PLQYs. Furthermore, the femtosecond transient absorption (fs-TA) spectra illustrated that the lifetimes were increased due to suppression of non-radiative process. In the solid state, the PLQY of **S3** was nearly 23 times higher than that of ligand **L**. This study demonstrates that the well-defined chromophore packing modes—particularly, stacking angle manipulation—have a pivotal impact on the luminescent behavior, which will further provide valuable insights for designing and preparing high-performance photoluminescent materials.

2 | Results and Discussion

In this study, the 180° di-Pt(II) acceptor **L** and a series of ditopic pyridyl ligands (**L1**–**L3**) with different lengths were designed and synthesized following the synthetic route shown in Schemes S1 and S2. All ligands were fully characterized by NMR, including ¹H, ¹³C, two-dimensional correlation spectroscopy (2D-COSY) (Figures S1–S13), as well as by matrix-assisted laser desorp-

tion/ionization time-of-flight mass spectrometry (MALDI-TOF MS) (Figures S32–S34).

The discrete complexes **S1**–**S3** were assembled with ligands **L1**–**L3** and **L** in a precise 1:1 stoichiometric ratio in dimethyl sulfoxide (DMSO) at 80 °C for 10 h (Figure 1a), which are in quantitative yields (95% for **S1**, 95% for **S2**, and 97% for **S3**). The formation of these discrete complexes was first confirmed by multinuclear NMR analysis, including ¹H, ³¹P, ¹³C, 2D-COSY, nuclear Overhauser effect spectroscopy (NOESY), and diffusion-ordered spectroscopy (DOSY) (Figure 1b–j, Figures S14–S31). In the ¹H NMR spectra of ligands **L1**, **L**, and **S1** (Figure 1b–d), the proton signals of **S1** corresponding to the pyridyl units and phenyl rings exhibited downfield shifts compared to **L1**, which can be attributed to the loss of electron density upon formation of Pt–N coordination bonds. The similar phenomenon was also observed in the ¹H NMR spectra of **S2** and **S3** (Figure 1e–j). Notably, the signals of the pyridyl and phenyl protons splitted into two sets upon metal coordination, which can be ascribed to the different chemical environments of phosphorus inside and outside due to the spatial arrangement in **S2** [69]. In the ³¹P NMR spectra of ligand **L** and complexes **S1**–**S3** (Figures S2, S15, S21, and S27), the di-Pt(II) acceptor **L** and complexes **S1** and **S3** exhibited a

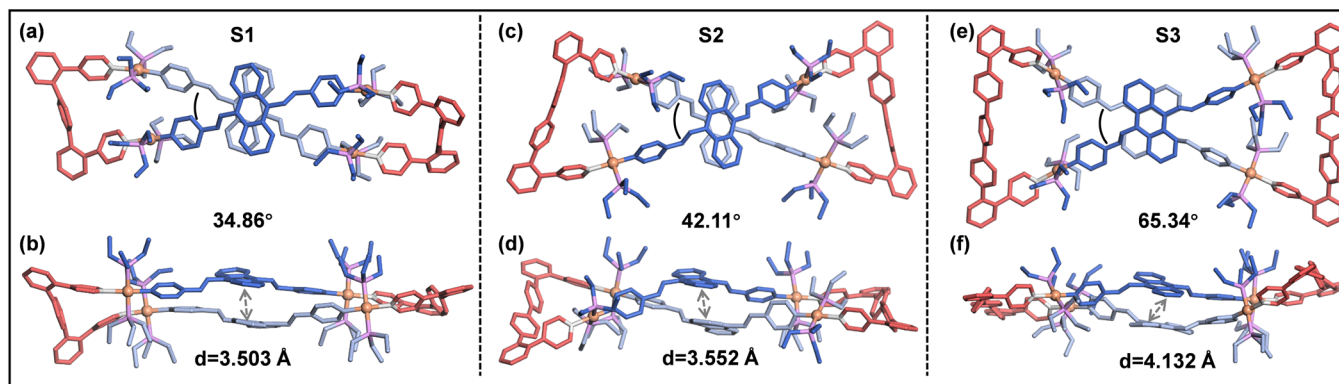


FIGURE 2 | Optimized structures of **S1** (a, b), **S2** (c, d), and **S3** (e, f) in the front view and side view by DFT calculation.

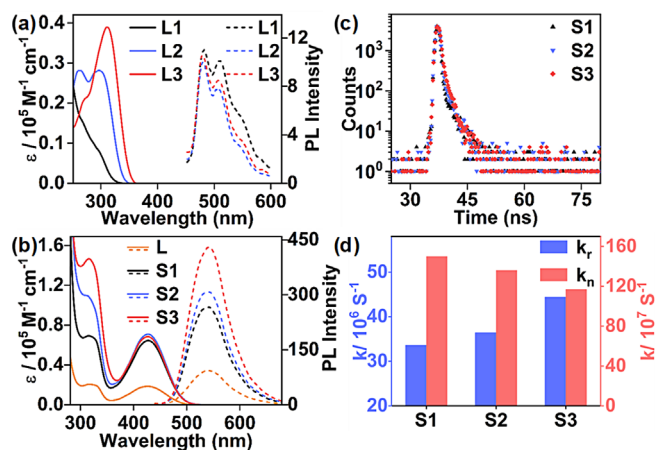


FIGURE 3 | UV-vis and PL spectra of **L1–L3** (a) in CHCl_3 and **L**, **S1–S3** (b) in DMSO; fluorescence decays for complexes **S1–S3** (c) in DMSO; the radiative and non-radiative rate constants of **S1–S3** (d) calculated from τ and Φ_F . k_r = radioactive decay rate (Φ_F/τ_s); k_{nr} = non-radiative decay rate ($1/\tau_s - k_r$) ($\lambda_{\text{ex}} = 429 \text{ nm}$, $c = 10.0 \mu\text{M}$).

single peak with concomitant ^{195}Pt satellites ($\delta = 18.60 \text{ ppm}$ for **L**, $\delta = 13.34 \text{ ppm}$ for **S1**, and $\delta = 13.04 \text{ ppm}$ for **S3**, respectively) (Figures S15 and S27) [70]. In contrast, **S2** exhibited a bimodal peak due to the presence of two conformations ($\delta = 13.37$ and 13.27 ppm), consistent with the phenomenon of ^1H NMR (Figure S21) [69]. DOSY spectra further confirmed discrete species, with single diffusion bands observed at $\log D = -10.36$, -10.38 , and -10.17 for **S1–S3**, respectively (Figures S19, S25, and S31). The experimental diameters calculated via the Stokes–Einstein equation, were 4.4 nm (**S1**), 4.6 nm (**S2**), and 3.0 nm (**S3**), respectively, which matched well with the sizes of the expected structures. In addition, electrospray ionization mass spectrometry (ESI-MS) was used to validate the molecular compositions. From ESI-MS spectra, peaks at $m/z = 813.3$, 851.3 , and 889.3 Da corresponding to $[\text{M-4OTf}]^{4+}$, which matched well with the simulated isotope patterns of supramolecules **S1–S3**, respectively (Figure 1k–m).

To further elucidate the structural details, all attempts to grow single crystals of **S1–S3** were unsuccessful and DFT calculations were carried out to optimize the ground state configurations of **S1–S3** (Figure 2a–f). DFT results revealed that **S1–S3** formed crossing X-dimer structures. The inter-dimer angles of anthracene motif of **S1–S3** increased from 34.86° to 65.34° , and

the distances between two anthracene motifs were increased from 3.503 to $4.132 \text{ \AA}$.

After confirming the composition and configuration of **S1–S3**, the photophysical properties of all ligands and complexes were investigated. The UV-vis absorption and fluorescence emission of ligands **L1–L3** were first measured in chloroform (Figure 3a). **L1–L3** exhibited two strong absorption bands at approximately 265 and 310 nm , corresponding to the π – π^* transitions of the pyridine unit and conjugated benzene rings, respectively. The fluorescence spectra of **L1–L3** showed a weak emission center at about 475 – 510 nm . The UV-vis absorption and photoluminescence emission spectra of the anthracene-based ligand **L** and complexes **S1–S3** in DMSO (Figure 3b) exhibited two strong absorption bands at 320 and 429 nm . Compared to **L**, the molar absorption coefficients of the anthracene center at 429 nm were significantly enhanced in **S1–S3**. As for the emission spectra of **S1–S3**, the emission bands were centered at 545 nm . Notably, the emission intensity and absolute quantum yields increased with the increasing intramolecular packing angles of anthracene motifs from **S1** to **S3**, which are higher than that of **L**. The PLQYs of **S1** (2.19%), **S2** (2.26%), and **S3** (3.67%) were increased, compared with that of **L** (2.02%) (Figure S44). Furthermore, the time-resolved decayed lifetime (τ_s), radiative decay rates (k_r), and non-radiative decay rates (k_{nr}) for **S1–S3** in DMSO were also investigated (Figure 3c,d, Table S1). The values of τ_s for **S1–S3** were also increased in order, 0.651 , 0.715 , and 0.826 ns , respectively. The values of k_r for **S1–S3** were calculated to be 3.36×10^7 , 3.64×10^7 , and $4.44 \times 10^7 \text{ s}^{-1}$, respectively. The values of k_{nr} were 1.50×10^9 , 1.36×10^9 , and $1.17 \times 10^9 \text{ s}^{-1}$ for **S1–S3**, respectively. The k_{nr}/k_r ratio for **S3**, exhibiting the largest stacking angle, was 26.35 , which was much smaller than that of **S1** (44.64), indicating that larger stacking angle more effectively suppressed the energy loss via non-radiative decay processes. In addition, low-temperature fluorescence spectra of **S1–S3** were performed by rapidly freezing the assemblies in DMSO at 77 K (Figure S43). Compared to emission intensities of the complexes at room temperature, the emission intensities were significantly enhanced at 77 K because the intramolecular rotation of anthracene motifs was profoundly limited in frozen state.

To gain deeper insight into the relationship between molecular conformation and luminescent properties, simplified-Tamndancoff approximation DFT (sTDA-DFT) calculations were employed to compute excited-state energy levels, and natural

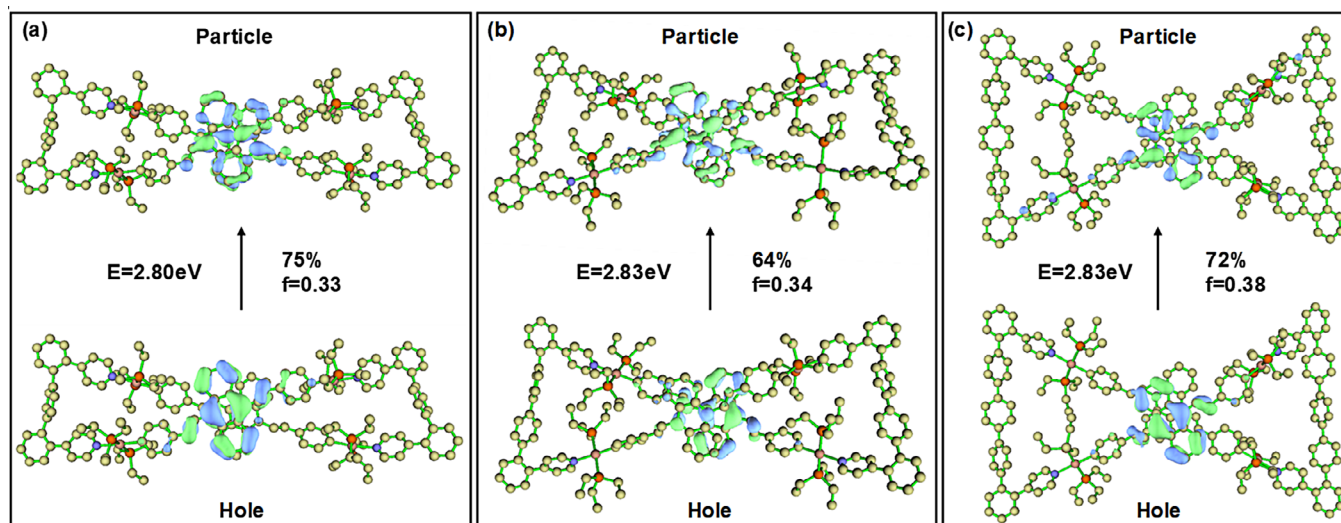


FIGURE 4 | Natural transition orbital (NTO) analysis of **S1** (a), **S2** (b), and **S3** (c) by sTDA-DFT calculations.

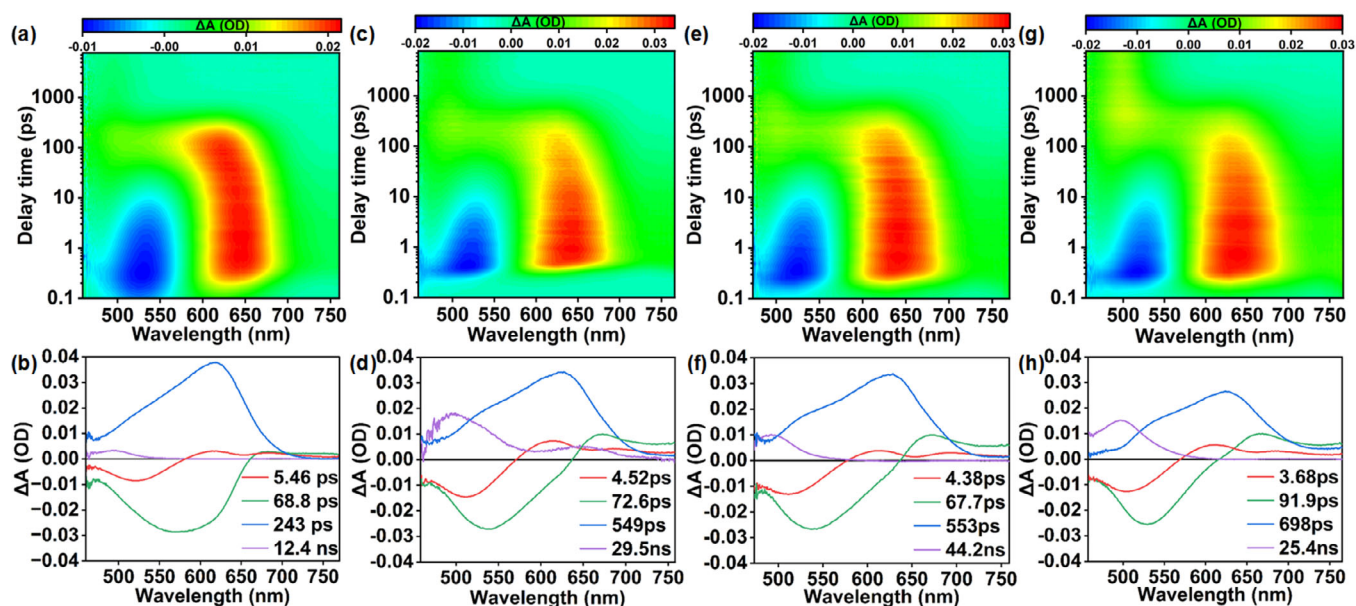


FIGURE 5 | Transient absorption (TA) spectra and the state-associated decay spectra (SADS) of the TA spectra of the DMSO solution of **L** (a, b), **S1** (c, d), **S2** (e, f), and **S3** (g, h) in DMSO.

transition orbital (NTO) analysis was used to study their electronic structures (Figure 4a–c). The sTDA-DFT results showed that as the stacking angle and distance of anthracene motifs in **S1–S3** increased, the S1–S0 energy gap rose from 2.80 eV (**S1**) to 2.83 eV (**S3**) and the radiative transition oscillator strength increased from 0.33 (**S1**) to 0.38 (**S3**), which are related to the gradually weakened intramolecular interaction. NTO analysis further revealed that the first excited states of **S1–S3** are all local excited states. In **S1** and **S2**, the NTO orbital pairs with the highest contributions to the first excited state (75% and 64%) were located on the anthracene rings of the two anthracene motifs. In contrast, in **S3**, the most contributing pair (72%) was distributed mainly on anthracene ring of one anthracene motif. The enhanced localization of excited-state wavefunctions from **S1** to **S3** suggested weakened interactions between anthracene

motifs, which correlated with the observed increase in radiative transition rates from **S1** to **S3** (Figure 3d).

To illustrate the excited-state dynamics, the fs-TA spectroscopy was further employed (Figure 5 and Figure S45). The fs-TA spectra of **L** and **S1–S3** were similar to that of anthracene motif. The decay-associated spectra (DAS) of the fs-TA data were fitted with singular value decomposition (SVD) and global fitting. In the state-associated decay spectra (SADS) of **L** and **S3**, the excited-state species A with a decay lifetime from 3.23 ps (**L**) to 4.38 ps (**S3**) were corresponding to the vertical excited state (VES) after Franck–Condon transition. As the excited-state molecules relaxed on the potential energy surface (PES), this species continued to form the excited-state species B with a lifetime from 62.9 ps (**L**) to 91.9 ps (**S3**). Compared to species A, the

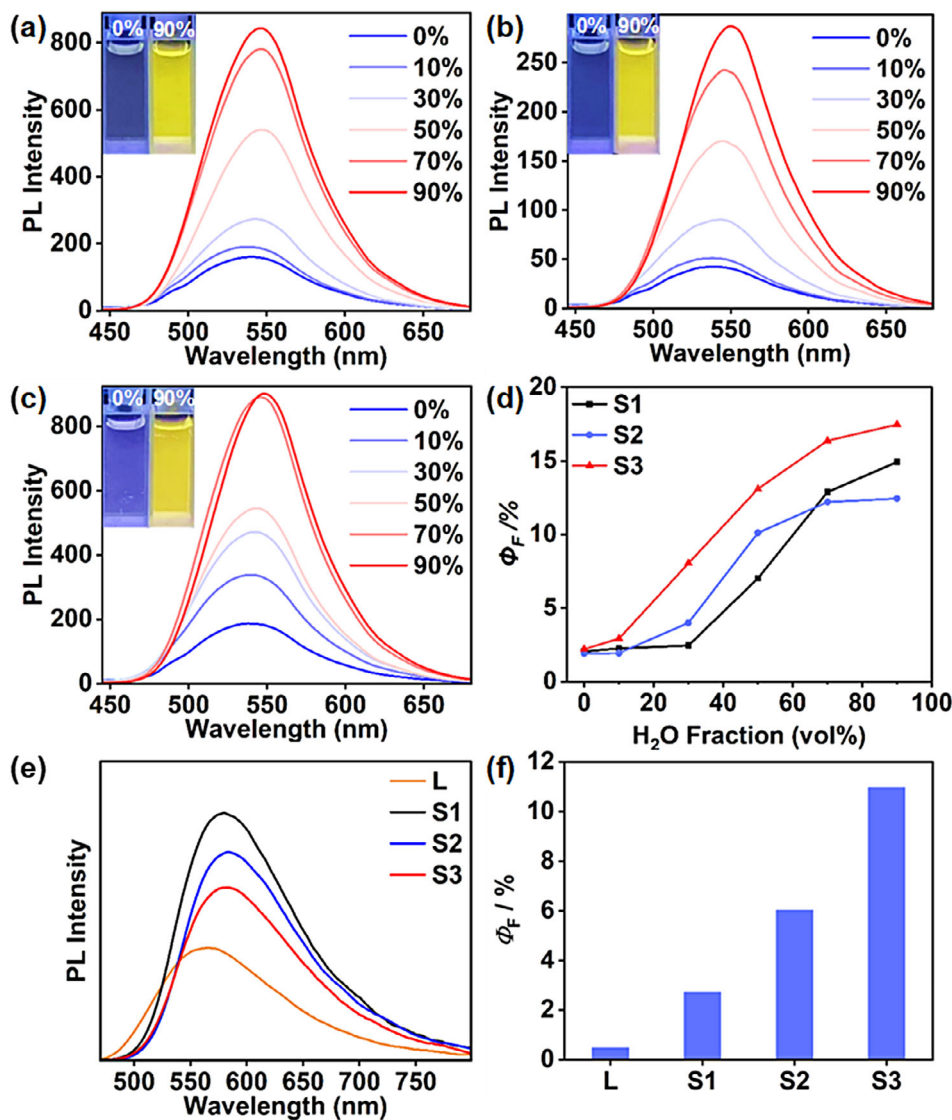


FIGURE 6 | PL spectra of **S1** (a), **S2** (b), and **S3** (c) recorded in DMSO/H₂O containing different H₂O fractions (the inserted pictures of **S1–S3** dissolved in DMSO/H₂O with 0% and 90% H₂O); PLQYs of **S1–S3** (d) versus increasing H₂O fractions in DMSO/H₂O; PL spectrum (e) and PLQYs (f) of **L**, **S1**, **S2**, and **S3** in solid state.

stimulated emission (SE) peak of species B was bathochromically shifted to 450–650 nm. Species B was further relaxed to form species C with a lifetime of 0.284–0.698 ns, resembling the fluorescence lifetime measured in the time-correlated single photon counting (TCSPC) spectra (Figure 3). Therefore, species C was attributed to the molecule's adiabatic excited state (AES; the local minimum point of the first excited-state PES), characterized by an excited-state absorption (ESA) peak covering 450–750 nm with the strong transition intensity that overlapped the SE signal. In addition, species D with a long decay lifetime (29.5–44.2 ns) was further formed with species C, which is a dark state with its ESA signal in the range of 480–550 nm. It was speculated to be the T1 state formed from the AES state (species C) through spin-orbit coupling due to the coordination of Pt atoms.

The lifetime of species A and species B showed no significant change from **L** to **S3**, as it is difficult to completely restrict

the PES relaxation of the VES by the conformational constraints on anthracene motifs. Due to the interactions between the intramolecular anthracene motifs, the lifetime of the AES state increased from 0.284 (**L**) to 0.549 ns (**S1**). As the stacking angles increased from **S1** to **S3**, the lifetime of the AES state increased from 0.549 (**S1**) to 0.689 ns (**S3**). The prolonged lifetime of the AES state was approximately in accordance with the results obtained from TCSPC, which indicated that the prolonged lifetime of **S3** may arise from its inhibited intersystem crossing to the triplet dark state (non-radiative transition), thereby leading to increased PLQYs of **S1–S3** as the stacking angles increased [71].

The fluorescence emission spectra of complexes **S1–S3** were recorded in DMSO/H₂O mixtures to study their emission behavior upon aggregation, with DMSO as a good solvent and H₂O as a poor solvent (Figure 6a–d and Figure S38). As the water fraction increased from 0% to 90%, the fluorescent intensities and PLQYs

of **S1–S3** gradually increased, reaching the maximum at 90% H₂O (Figure 6a–d). In the aggregated state, vibrational motions of the benzene rings in anthracene motif were restricted, which enhanced the fluorescent intensity of **S1–S3**. In addition, the fluorescent emission of ligand **L** and complexes **S1–S3** in the solid state was further explored. Compared to ligand **L**, the emission spectra of **S1–S3** were redshifted by about 15–20 nm, possibly due to the molecular stacking after coordination (Figure 6e). Notably, the PLQYs of **S1–S3** in the solid-state were 2.72%, 6.03%, and 10.97%, respectively (Figure 6f). The PLQY value of **S3** significantly exceeded nearly 23-fold that of **L** (0.48%). These results further demonstrated that the excellent luminescent properties of **S3** are closely related with the largest stacking angle.

3 | Conclusion

In summary, three figure-eight supramolecules **S1–S3** were constructed by coordinating the anthracene-based 180° di-Pt(II) acceptor **L** with ditopic pyridyl ligands **L1–L3** varying in different lengths, respectively. The variation in ligand length enabled the regulation of intramolecular anthracene stacking angles in the assembled structures. The fluorescent intensity and PLQYs in the solution state were enhanced with increasing stacking angles of chromophore from **S1** to **S3**. DFT calculation revealed that the interactions of intramolecular anthracene motifs decreased from **S1** to **S3**, which led radiative transition rate gradually increasing and further enhanced the fluorescence intensity and PLQYs. The fs-TA spectra further demonstrated that the lifetime was increased due to suppression of the non-radiative process, which revealed the relationship between stacking angles and the luminescent properties of supramolecules. Moreover, the PLQY of **S3** significantly exceeded nearly 23-fold that of **L** (0.48%) in solid states. This work presents a novel strategy for tailoring the chromophore stacking angles to effectively tune the luminescent properties.

Acknowledgments

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting Information file 1: agt270198-sup-0001-SupMat.pdf